Roll No.: DL.AL.U4AID24135

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2025, Third Semester

Artificial Intelligence & Data Science

23AID113 Elements of Computing 2

Duration: Two hours **Maximum:** 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO1	Develop and execute programs in low-level languages such as Hack machine language and assembly language.
CO2	Create virtual machine (VM) specifications and generate VM code for programs written in high-level and assembly languages.
CO3	Design and implement translators/compilers that convert high-level constructs into VM or assembly code, and validate the generated code through testing and debugging.
CO4	Develop programs in the object-based language Jack and execute experiments related to basic concepts and functions of operating systems and compilers.

Answer all questions

		Marks	CO	BTL
1.	Answer all the questions: (a) What is a machine instruction? (b) Why are registers important in assembly language programming? Explain their purpose in executing instructions in short. (c) What do you understand by a segmented memory model? Explain its purpose in computer systems in short. (d) Explain what an assembler does, and why it's required. (e) Define addressing modes. How do they help in computer architecture?	[5×2]	1	1
2	Answer all the questions: a) Define Stack Arithmetic with reference to the Virtual Machine. b) State any four program control instructions in a VM. c) How does the Jack language integrate with the VM layer? d) Outline any two benefits of high-level programming using Jack. e) Describe the implementation steps for the basic VM arithmetic command.	[5×2]	2	1
3	Explain the flow from high-level Jack code to machine code (binary code).	[10]	3	2

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4	Explain with an example the difference between assembly and machine language, highlighting syntax and conversion steps.	[10]	1	4
5	Briefly explain the Virtual Machine (VM). Discuss in detail the VM Arithmetic commands and the VM Program Control commands.	[10]	2	4

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	20	BTL 1	20
CO2	20	BTL 2	10
CO3	10	BTL 3	
CO4		BTL 4	20
CO5		BTL 5	